Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/29/2020
PI Name:	Chiu, Charles M.D., Ph.D.		
Project Title:	In-Flight Metagenomic Monitoring of Infections and Associated Host Responses in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	charles.chiu@ucsf.edu	**Fax:**	FY 415-889-6441
PI Organization Type:	UNIVERSITY	**Phone:**	415-218-9083
Organization Name:	University of California, San Francisco		
PI Address 1:	Department of Laboratory Medicine and Medicine / Infectious Diseases		
PI Address 2:	185 Berry St Box #0134		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94107-5705	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
Start Date:	10/01/2017	**End Date:**	09/30/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 (original end date was 9/30/2022) per E. Urquieta/TRISH (Ed., 6/2/19) NOTE: End date changed to 6/30/2019 (original end date was 9/30/2022) per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Juran, Cassandra Ph.D. (NASA Ames Research Center) Lupisella, Mark Ph.D. (NASA Goddard Space Flight Center) Smith, David Ph.D. (NASA Ames Research Center) Almeida, Eduardo Ph.D. (NASA Ames Research Center) Mason, Christopher Ph.D. (Weill Medical College of Cornell University)		
Grant/Contract No.:	NNX16AO69A-T0102		
Performance Goal No.:			
Performance Goal Text:			

REPORT SUBMITTED DECEMBER 2019
Flight Definition Phase

(1) Engagement with Johnson Space Center International Space Station Medical Project (JSC ISSMP) regarding aspects of the proposed astronaut host response and microbiome study, including (1) obtaining human subjects approvals, (2) flight plan, (3) choice of kits, and (4) sample collection scheduling. (2) Testing 10 de-identified blood samples on the nanopore sequencer to assess feasibility of RNA-Seq analysis. (3) Engagement with ISS computing group on installation and deployment of nanopore analysis software on flight-loaded HP ZBook G2 laptops (Stephen Hunter).

Full Project Specific Aim 1 (Years 1-3): Develop an infrastructure for real-time, sequencing based monitoring of infections and host responses in astronauts aboard the ISS. 1A: Test and validate a flexible, robust, and automated protocol for nanopore sequencing from clinical and environmental samples. The protocol will be customized for metagenomic identification of pathogens and characterization of human host responses by RNA sequencing. 1B: Develop and test user-friendly software for real-time nanopore sequencing analysis on a laptop. Crucially, running the software will not require internet connectivity or user bioinformatics expertise. We will deploy these laptops aboard the ISS for field testing. 1C: Benchmark the protocol on Earth using clinical samples and standardized controls. We will analyze well-characterized patient samples (with complete metadata) at University of California San Francisco (UCSF) standardized commercial and in-house (UCSF metagenomic sequencing assay) controls. 1D: Benchmark the protocol on Earth using available clinical samples collected in-flight from the NASA TWINS study or Microbial Tracking-2 study.

Specific Aim 2 (Years 3 and 4): Demonstrate the utility of the nanopore sequencing platform for 2 specific use cases aboard the ISS. 2A: In-flight analysis of the stool microbiome of astronauts and environmental microbiome of the ISS. We will collect samples longitudinally over several time points before, during, and after flight, to compare the astronaut stool microbiome with the environmental microbiome (swabs of the waste and hygiene compartment in Node 3, dining table in Node 1, and crew sleeping quarters in Node 2). With multiple collection time points, we will identify patterns of transmission and colonization of these organisms in the environment of the ISS. 2B: In-flight longitudinal monitoring of the plasma (blood) virome and associated host responses of astronauts. We will perform longitudinal RNA sequencing (RNA-Seq) and metagenomic DNA/RNA sequencing and TCore Proposal: NNJ16ZSA001N-TRIRT-Chiu PI: Chiu, Charles, MD/PhD (University of California, San Francisco) correlate reactivation of viral infections (e.g., herpesviruses) with patterns of host response.

Our proposed project will span technology readiness levels 5 - 8, and will involve 2 stages. The first stage will be Earth-based research. These efforts will involve optimization of clinical sample preparation and nanopore sequencing analysis as well as software development and validation for local, real-time computational analysis on laptop computers. In the second stage, we will collect clinical samples (whole blood, plasma, and respiratory secretions) and associated environmental samples aboard the ISS and analyze them on board by real-time nanopore sequencing. Sequencing analyses will involve primarily agnostic omics (metagenomics, 16S microbiome, RNASeq) techniques, with the use of qRT-PCR or library enrichment using oligonucleotide probes of a multiplexed panel of specific biomarkers for validation. A subset of samples, following Wetlab-2 nucleic acid extraction and stabilization (both pre- and postamplificatoin), will be returned to the Earth for validation using gold-standard instruments.

REPORT SUBMITTED FALL 2018

Last year’s successful International Space Station (ISS) demonstration of sample processing, qRT-PCR, and DNA sequencing makes the real-time genomic monitoring of astronaut health during spaceflight possible. The use of omics methods can enable in-depth identification of the spectrum and diversity of microorganisms that cause acute infectious diseases in humans. Additionally, omics approaches during spaceflight could dynamically assess human host responses to bacterial colonization and/or viral reactivation.

We have pioneered the use of metagenomic next-generation sequencing (mNGS) for detection of all potential pathogens – viruses, bacteria, fungi, and parasites – in a single clinical assay (Greninger, et al., 2015), using this approach to save the life of a critically-ill 14-year boy with neuroleptospirosis (Wilson, et al., 2014). We have also shown the utility of omics for assessing human host responses by transcriptome profiling (RNA-Seq), leveraging this approach to identify a distinct signature for Lyme disease (Bouquet, et al., 2016). On Earth, failure to diagnose acute illnesses in a timely fashion directly contributes to poor patient outcomes, rapid spread of outbreaks, and emergence of antibiotic resistance -- issues that would only be magnified in space and potentially affect all crew members within the closed environment of a spacecraft.

Task Description:

Here we propose a multidisciplinary effort to develop and validate foundational infrastructure for routine in-flight metagenomic monitoring of potential infections and host responses to infection in astronauts. Our two specific aims are as follows:

1. Establish an infrastructure for real-time, sequencing-based monitoring of infections and host responses in astronauts aboard the ISS.

2. Demonstrate the utility of our nanopore-sequencing based platform for 2 specific use cases, as follows:

a. In-flight analyses of the respiratory bacterial microbiome of pooled oropharyngeal/nasopharyngeal swabs from astronauts, and comparison to the environmental microbiome (e.g., toilet, dining table, sleeping quarters).

b. In-flight longitudinal monitoring of the plasma (blood) virome in astronauts over time, with associated analyses of human host responses to viral reactivation.

This is a collaborative effort involving NASA scientists and clinical/academic faculty from University of San Francisco and Weill-Cornell Medicine. We will leverage pre-existing sample collection, nucleic acid purification, qRT-PCR and nanopore sequencing approaches developed for use on the ISS, coupled with our broad expertise in nanopore sequencing, computational methods, microbiology, and infectious diseases. Our collaboration recently resulted in both molecular and genomic sequencing data generated in-flight aboard the ISS (Castro-Wallace et al., 2017; Almeida et al., 2017). The proposed project will span technology readiness levels 5 – 8.

Our proposal will involve 2 stages. The first stage will be Earth-based research. These efforts will involve preparation of clinical samples and libraries to be sent to the ISS for nanopore sequencing analysis and software development and validation for local, real-time computational analysis using on-board laptop computers. In the second stage, we will

collect clinical samples (whole blood, plasma, and respiratory secretions) and associated environmental samples aboard the ISS and analyze them by real-time nanopore sequencing. Sequencing analyses will involve primarily agnostic omics (metagenomics, 16S microbiome, RNA-Seq) techniques, with the use of qRT-PCR for validation.

Given the focus on deep space missions including a planned manned trip to Mars, it is imperative that astronauts be equipped with omics capabilities to monitor infection during spaceflight in real-time. Our project proposes to develop a blueprint for incorporating microbial and human gene expression-based sequencing as routine on-board tools for monitoring astronaut health.

References

Greninger AL, Messacar K, Dunnebacke T, Naccache SN, Federman S, Bouquet J, Mirsky D, Nomura Y, Yagi S, Glaser C, Vollmer M, Press CA, Kleinschmidt-DeMasters BK, Dominguez SR, and Chiu CY. (2015) Clinical metagenomic identification of Balamuthia mandrillaris encephalitis and assembly of the draft genome: the continuing case for reference genome sequencing. Genome Medicine 7(1):113.

Greninger AL, Naccache SN, Messacar K, Clayton A, Yu G, Somasekar S., Federman S, Stryke D, Anderson C, Yagi S, Messenger S, Wadford D, Xia D, Watt JP, Van Haren K, Dominguez SR, Glaser C, Aldrovandi G, and Chiu CY. (2015) A novel outbreak enterovirus D68 strain associated with acute flaccid myelitis cases in the United States from 2012-2014: a retrospective cohort study. Lancet Infectious Diseases 15(6):671-682.

Wilson MR, Naccache SN, Samayoa E, Biagtan M, Bashir H, Yu G, et al. (2014) Actionable diagnosis of neuroleptospirosis by next-generation sequencing. The New England Journal of Medicine. 2014;370(25):2408-17.

Bouquet J, Soloski MJ, Swei A, Cheadle C, Federman S, Billaud, J-N, Rebman AW, Kabre B, Halpert R, Boorgula M, Aucott J, and Chiu CY. (2016) Longitudinal transcriptome analysis reveals a sustained differential gene expression signature in patients treated for acute Lyme disease. mBio 7(1):e00100-16.

Castro-Wallace S, Chiu C, John K, Stahl S, Rubins K, A.B.R. M, et al. (2017) Nanopore DNA sequencing and genome assembly on the international space station. bioRxiv 077651.

Parra M, Jung J, Boone TD, Tran L, Blaber EA, Brown M, Chin M, Chinn T, Cohen J, Doebler R, Hoang D, Hyde E, Lera M, Luzod LT, Mallinson M, Marcu O, Mohamedaly Y, Ricco AJ, Rubins K, Sgarlato GD, Talavera RO, Tong P, Uribe E, Williams J, Wu D, Yousuf R, Richey CS, Schonfeld J, Almeida EAC. (2017) Microgravity validation of a novel system for RNA isolation and multiplex quantitative real time PCR analysis of gene expression on the International Space Station. PLoS One. 2017 Sep 6;12(9):e0183480.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

REPORT SUBMITTED DECEMBER 2019
Given the recent focus on planned missions to the moon and Mars, our project proposes to equip astronauts with omics capabilities to monitor infection and host responses during spaceflight in real-time. The sequencing protocol provided will enable microbial and human gene expression sequencing to be performed as routine on-board tools for monitoring astronaut health, especially on deep space missions. The protocol will also be of benefit to Earth-based missions as it can be readily performed in remote and low-resource settings.

REPORT SUBMITTED FALL 2018

The successful demonstration of DNA sequencing in microgravity using the MinION nanopore sequencer involving most of our research team (Castro-Wallace, et al., 2017, Scientific Reports, 7:18022) has enabled the prospect of real-time metagenomic monitoring of astronaut health during spaceflight. Having "omics" capabilities on-board introduces a number of immediate and potentially life-saving applications in the fields of microbiology and infectious diseases including (1) rapid diagnosis of infections in astronauts by metagenomic analysis of clinical samples such as blood, respiratory secretions, or stool; (2) microbiome screening of environmental, food, water, and human samples to identify microbial pathogens that may threaten the health of astronauts; and (3) longitudinal monitoring of the astronaut microbiome and virome over time in parallel with associated human host biomarkers that may be diagnostic for illnesses, whether infectious or noninfectious. With acute infectious diseases, time is of the essence, and return of clinical and/or environmental samples to ground-based laboratories for analysis is not a viable option, particularly with planned missions beyond low Earth orbit that are intended to be long duration, including to Mars and beyond. Thus, implementation of these "omics" technologies is critically needed for continual, real-time surveillance of astronaut health. As our team has previously delivered in the individual technical aspects of generating omics results in space, we are now ready to use the International Space Station for showing operational feasibility of a fully integrated omics workflow. Our Translational Research Institute (TRI) project will optimize omics microgravity procedures and produce sequencing results that can be analyzed on-orbit, in real-time, using only a laptop computer.

REPORT SUBMITTED DECEMBER 2019
In-Flight Metagenomic Monitoring of Infections and Associated Host Responses in Astronauts

The successful demonstration of DNA sequencing aboard the International Space Station (ISS) in 2017 introduced the possibility of real-time genomic monitoring of astronaut health during the planned manned missions to the Moon and Mars This project aims to establish an infrastructure for routine in-flight metagenomic monitoring of potential infections and host responses to infection in astronauts. We propose to collect a number of different clinical biosamples from astronaut crew members pre-, during, and post-flight, including whole blood, plasma, nasal swab, stool swab, and skin swabs, as well as environmental swabs from the ISS (astronauts will collect biosamples from themselves during flight). Using established methods, we will guide astronauts in processing and sequencing their clinical samples in-flight, and analyze the sequencing data in real-time on a laptop housing the SURPIrt bioinformatics software developed at University of California, San Francisco. Astronaut crew will demonstrate in this proof-of-principle project that it will be possible in-flight to (1) monitor viral infections in their blood and nasal swabs over time and associated human host responses and (2) monitor the microbiome (bacteria) of the astronauts from nasal, skin, and stool swab samples, and compare this to the local environment. Our proposal will involve 2 stages. The first stage (Years 1-2) will be Earth-based research, where we develop, adapt, and optimize the sample processing and sequencing protocols for analyses aboard

Task Progress:	the ISS. In the second stage (Years 3-4), we will engage astronauts to perform the sample collection and sequencing analyses in-flight. In Year 1 of the project, we have modified the previously flight-validated SPM (sample preparation module) for nucleic acid extraction for compatibility with a variety of clinical and environmental sample types as proposed in this project (plasma, whole blood, nasal swab, stool swab, environmental swab, and skin swabs). Ground-based testing using modified portable versions of the SPM has been performed on whole blood and bone marrow cell cultures to date. We have also developed and validated on the ground streamlined protocols for performing the 3 key library preparation steps prior to nanopore sequencing: (i) metagenomic DNA, (ii) RNA-Seq (RNA-based human host gene expression), (iii) 16S ribosomal DNA. We have modified existing laptop-based software for automated analysis of nanopore metagenomic sequencing data (SURPIrt) aboard the ISS. We have submitted an Institutional Review Board (IRB) protocol in conjunction with ISSMP (International Space Station Medical Projects) for human sample collection and processing aboard the ISS in preparation for in-flight project launch in late 2019 / early 2020. It is imperative that astronauts be equipped with omics capabilities to monitor infection during spaceflight in real-time. Our project thus proposes to develop a blueprint for incorporating microbial and human gene expression-based sequencing as routine on-board tools for monitoring astronaut health. In Year 2 of the project, we have completed all key milestones needed for initiation of in-flight testing. First, we have validated the 2-step, 1-tube RNA extraction and library preparation protocol on a range of clinical samples, including in low-resource environments. Second, we have confirmed that SURPIrt pipeline can process data directly from the MinIT, which can make the computational steps easy and seamless for crew, as well as published our computational methods. Third, we have coordinated with JPL (Jet Propulsion Laboratory) and CASIS (Center for the Advancement of Science in Space) to reduce crew time and ensure any extra needed support is available. Fourth, we have commitments for matching funds from several industry and Foundation partners, to ensure success and reduce TRISH (Translational Research Institute for Space Health) costs. Finally, we have continually met as a team to coordinate efforts and we are all fully dedicated to the next steps of ISS deployment in 2019-2020. REPORT SUBMITTED FALL 2018 The successful demonstration of DNA sequencing aboard the International Space Station (ISS) in 2017 introduced the possibility of real-time genomic monitoring of astronaut health during the planned manned missions to the Moon and Mars. This project aims to establish an infrastructure for routine in-flight metagenomic monitoring of potential infections and host responses to infection in astronauts. We propose to collect a number of different clinical biosamples from astronaut crew members pre-, during, and post-flight, including whole blood, plasma, nasal swab, stool swab, and skin swabs, as well as environmental swabs from the ISS (astronauts will collect biosamples from themselves during flight). Using established methods, we will guide astronauts in processing and sequencing their clinical samples in-flight, and analyze the sequencing data in real-time on a laptop housing the SURPIrt bioinformatics software developed at University of California, San Francisco. Astronaut crew will demonstrate in this proof-of-principle project that it will be possible in-flight to (1) monitor viral infections in their blood and nasal swabs over time and associated human host responses and (2) monitor the "microbiome" (bacteria) of the astronauts from nasal, skin, and stool swab samples, and compare this to the local environment. Our proposal will involve 2 stages. The first stage (Years 1-2) will be Earth-based research, where we develop, adapt, and optimize the sample processing and sequencing protocols for analyses aboard the ISS. In the second stage (Years 3-4), we will engage astronauts to perform the sample collection and sequencing analyses in-flight. In Year 1 of this project, we have modified the previously flight-validated SPM (sample preparation module) for nucleic acid extraction for compatibility with a variety of clinical and environmental sample types as proposed in this project (plasma, whole blood, nasal swab, stool swab, environmental swab, and skin swabs). Ground-based testing using modified portable versions of the SPM has been performed on whole blood and bone marrow cell cultures to date. We have also developed and validated on the ground streamlined protocols for performing the 3 key library preparation steps prior to nanopore sequencing: (i) metagenomic DNA, (ii) RNA-Seq (RNA-based human host gene expression), and (iii) 16S ribosomal DNA. We have modified existing laptop-based software for automated analysis of nanopore metagenomic sequencing data (SURPIrt) aboard the ISS. We have submitted an Institutional Review Board (IRB) protocol in conjunction with ISSMP (International Space Station Medical Projects) for human sample collection and processing aboard the ISS in preparation for in-flight project launch in late 2019 / early 2020. It is imperative that astronauts be equipped with omics capabilities to monitor infection during spaceflight in real-time. Our project thus proposes to develop a blueprint for incorporating microbial and human gene expression-based sequencing as routine on-board tools for monitoring astronaut health.
Bibliography Type:	Description: (Last Updated:)